WASP-50b

R-1464

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_211024` · created 2026-08-01T11:08:29+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459511.8127 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.016
Transit depth (Rp/Rs)^2	2.69 +/- 0.52 [%]
Orbital Inclination (inc)	83.96 +/- 0.74
Airmass coefficient 1 (a1)	1.106 +/- 0.017
Airmass coefficient 2 (a2)	-0.068 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	None
Optimal Aperture	4.04
Optimal Annulus	6.68
Transit Duration (day)	0.0697 +/- 0.0082

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.016 vs 0.1404 (deviation 1.48σ)
Duration fit vs published	0.0697 d vs 0.0754 d (deviation 0.7σ)
Mid-time O–C	3.2 min
Depth SNR	10.2
Residual scatter	1.51 %

Light curve

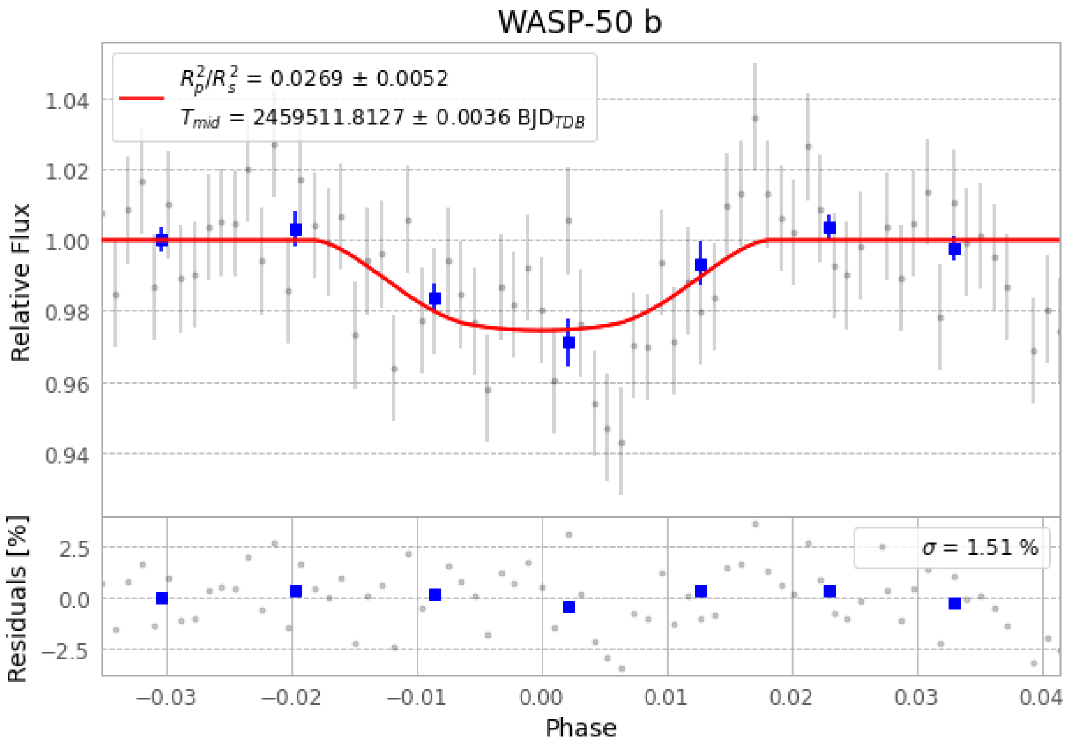

Figure 1 — FinalLightCurve_WASP-50 b_2021-10-23.png (SHA-256 38b342b21180cb77…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 68], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 fd92aeb3e29ddcc4…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

73 FITS frames (48 MB), `WASP-50211024054212.FITS` → `WASP-50211024091812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-211024.log (87ebde1431ac…), FinalLightCurve_WASP-50 b_2021-10-23.png (38b342b21180…), FinalLightCurve_WASP-50 b_2021-10-23.csv (0f36bcf81b7f…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)